MR Credit: Construction and Demolition Waste Management

1–2 Points

This credit applies to

- BD+C: New Construction
- BD+C: Core & Shell
- BD+C: Schools
- BD+C: Retail
- BD+C: Data Centers
- BD+C: Warehouses & Distribution Centers
- BD+C: Hospitality
- BD+C: Healthcare

INTENT

To reduce construction and demolition waste disposed of in landfills and incineration facilities through waste prevention and by reusing, recovering, and recycling materials, and conserving resources for future generations. To delay the need for new landfill facilities that are often located in frontline communities and create green jobs and materials markets for building construction services.

REQUIREMENTS – Waste Management Plan and Report

All projects must develop and implement a construction and demolition waste management plan:

- Identify strategies to reduce the generation of waste during project design and construction.
- Establish waste diversion goals for the project by identifying the materials (both structural and nonstructural) targeted for diversion.
- Describe the diversion strategies planned for the project. Describe where materials will be taken including expected diversion rates for each material.

Provide a final waste management report detailing all waste generated, including disposal and diversion rates for the project. Calculations can be by weight or volume but must be consistent throughout. Exclude excavated soil and land-clearing debris from calculations. Include materials destined for alternative daily cover (ADC) in calculations as waste (not diversion). Any materials sent to a commingled recycling facility must take the facility average recycling rate and include ADC as waste.

Option 1. Diversion (1 point)

Follow the Waste Management Plan and divert at least **50%** of the total construction and demolition materials from landfills and incineration facilities.

Option 2. Waste Prevention (2 points)

Follow the Waste Management Plan and prevent waste through reuse and source reduction design strategies. Salvage or recycle at least 50% of demolition debris and utilize waste minimizing design strategies and construction techniques for new construction elements.

Points are awarded as follows:

- Divert at least 50% of all demolition waste, if any. Generate less than **10 lbs./ft^2 (50 kg/m^2)** of waste materials from all new construction activities (2 points).

GUIDANCE – Behind the Intent

Beta Update

The requirement for diversion via multiple material streams in LEED has been removed. Project teams now have more flexible ways to prevent and divert waste. This structure rewards the hierarchy of first reduce, reuse, and then recycle. Additional updates clarify when waste-to-energy can count as diversion for international projects.

GUIDANCE – Step-by-Step

Option 1: Diversion

After exploring source reduction strategies, determine strategies for on-site and off-site waste collection during construction. Projects may use a combination of on-site separation and commingled collection. Strategies include:

- Stage collection bins onsite to correspond with construction phases and contractor schedules.
- Source separated materials tend to have higher diversion rates than mixed-waste processing.
- Donate surplus or salvaged materials.
- Participate in manufacturer take-back and recycling programs for ceiling tiles, furniture, or flooring.
- Incineration of some C&D; materials (other than wood) may be considered diversion for international projects only if reuse and recycling methods are not readily available.

Option 2: Waste Prevention

Track the amount of waste generated during demolition work separately from new construction activities. For Option 2, the project must demonstrate at least 50% of demolition waste was diverted (ADC does not count as diversion). Source reduction strategies include:

- Reuse buildings or building components (materials reused on-site are excluded from waste generation calculations).
- Reuse existing materials and components, design for modular construction sizes, specify reduced packaging.
- Design for industry-standard measurements, eliminate unnecessary finishes, off-site prefabrication.
- Work with subcontractors to eliminate or minimize packaging waste.

GUIDANCE – Waste-to-Energy

Combustion of wood materials (wood-derived fuel or biomass) is acceptable diversion for all projects and is not considered waste-to-energy. Forms of waste-to-energy (other than wood) are not acceptable diversion for US projects. For international projects, waste-to-energy may be eligible under the Alternative Compliance Path if European Commission Waste Framework Directive 2008/98/EC and Waste Incineration Directive 2000/76/EC are followed.

EXEMPLARY PERFORMANCE

- Achieve Option 1 and Option 2.
- OR: Divert 50% or more of all waste and utilize certified recycling facilities for all commingled waste. Presently, the Recycling Certification Institute's Certification of Real Rates (CORR) protocol meets certified facility requirements. Find facilities at: www.recyclingcertification.org/certified-facilities

REQUIRED DOCUMENTATION

- A compliant Construction and Demolition Waste Management Plan.
- A narrative for waste prevention/design strategies and calculation of total waste per area (if pursuing Waste Prevention points).
- MR construction and demolition waste management calculator demonstrating total and diverted waste amounts and documentation of recycling rates for commingled facilities.
- International projects: justification narrative for WTE strategy and documentation of WTE facilities adhering to EN standards (if applicable).

CONNECTION TO ONGOING PERFORMANCE

- LEED O+M MR prerequisite Waste Performance: A similar measure is required for the O+M v4.1 rating system and is a strategy that can help achieve the MR prerequisite: Waste Performance.